Mode-Conditional Assembly: Compositional Set Inference for Class-Agnostic Few-Shot Logical Anomaly Detection

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Abstract

Logical anomalies—a missing part, an extra part, or a part displaced from its legal relative position—are diagnosed by asking whether a test image is a legal composition *of parts. Under the class-agnostic few-shot protocol (a single generic prompt "part", no category name, $K \leq 4$ normal support images), the strongest existing system, ObjectCore, implicitly makes two unverified assumptions: that the K support images share one layout, and that parts may be freely recombined across support images. We show both assumptions fail in practice and propose* ***MCA*** *(Mode-Conditional Assembly), a non-parametric compositional set-inference framework. Given $K \leq 4$ support images, MCA enumerates all Bell(K) layout partitions, selects and averages over them by leave-one-image predictive NLL, and scores a query by conjugate posterior-predictive assembly cost: a Gamma–Poisson negative-binomial parts-inventory term and per-dimension 1D Normal–Inverse–Gamma Student-t pairwise-relation terms, with leave-one-support-out soft voting gating weak relations. Single-mode episodes degenerate exactly to nearest-support matching, so MCA cannot lose the existing lower bound on homogeneous support sets. We benchmark on the class-agnostic 4-shot MVTec LOCO and CAD-SD protocols of ObjectCore [****[TODO: main-table numbers pending full runs; ETA 2026-10-30]****].*

1. Introduction

Vision foundation models have pushed *structural* few-shot anomaly detection close to practical saturation. Starting from WinCLIP [23], which ensembled thousands of manually designed text templates over CLIP, the field quickly automated prompt engineering [29] and object-agnostic prompt learning [11, 42]. A parallel line then abandoned text altogether: frozen self-supervised backbones such as DINOv2 turned few-shot AD into a pure retrieval problem, raising one-shot image-level AUROC on MVTec-AD from 93.1% to 96.6% with AnomalyDINO [12], and further to 97.4% with a nearest-neighbour framework that fuses multi-layer foundation features [39]. Most recently, SubspaceAD showed that even a single PCA subspace fitted to DINOv2 patch features of k normal images suffices, reaching 97.1% one-shot AUROC on MVTec-AD without any training, memory bank, or prompt tuning [27]. Together, these results suggest that, for *structural* defects such as scratches, stains, and broken textures, expressive frozen features are by themselves nearly enough.

This saturation is restricted to appearance defects. The patch-based assumption—that a normal image is merely a collection of normal local textures—breaks down when the anomaly lies in *composition*: a missing component, a misplaced part, or an illegal co-occurrence. SubspaceAD's own authors state this boundary directly: patch-based methods "focus on local appearance and therefore do not explicitly account for logical or structural anomalies, such as missing or misplaced components," and "the model lacks the semantic, object-level understanding to know that a component should be present in that specific location" [27]. Vision-only retrieval methods inherit the same limitation: they compare query patches against stored normal patches without ever asking whether the detected parts form a legal assembly [12, 39]. Text-prompt methods do not close the gap either—they require per-class category names in the prompts [23, 29] or learn generic normal/abnormal semantics on auxiliary datasets rather than from the few target images themselves [42]. On the logical-anomaly benchmark MVTec LOCO, one-shot methods grounded in this paradigm remain near or below 68% AUROC even with large VLM backends (e.g. AnomalyMoE at 67.9% one-shot LOCO, as compiled by [3]).

We study the hardest protocol for logical anomaly detection: *class-agnostic few-shot*. The detector receives $K \leq 4$ normal images of an unseen category, a single generic grounding prompt ("part"), and no category name. This is the protocol introduced by ObjectCore [14] on MVTec LOCO [8] and CAD-SD [21]. Reading the recent main-conference record, every few-shot anomaly detection sys-

tem that entered CVPR/ICCV/ECCV/WACV in the last three years—WinCLIP [23], PromptAD [29], Anomaly-DINO [12], MuSc [28], SubspaceAD [27], SALAD [13]—did so by winning on a *newly defined protocol*, not by re-scoring a saturated one. We therefore position this work as a system paper with a clear leaderboard: beat the class-agnostic 4-shot image-AUROC of ObjectCore [14] on MVTec LOCO (80.8 ± 1.3) and treat CAD-SD (96.5 ± 0.6) as a point-estimate degeneration check. We do not compete on full-shot LOCO (SALAD [13] already reports 96.1 with per-category training) or on structural few-shot (already saturated by frozen-feature methods); MVTec AD / VisA appear only as an auxiliary sanity check that our structural branch does not collapse.

Two structural failures of the current best system. ObjectCore detects logical anomalies by retrieving the *single* most similar support image (by a global average patch feature f_s: "the most similar O_s is retrieved") and matching the test image's parts against that one image via Hungarian assignment. This embeds two assumptions.

(F1) One-layout assumption. It assumes all K support images share one layout. In real categories they do not: a breakfast box may pack the contents in several legal arrangements, and a cable splice may be wired several legal ways. When support images span $M > 1$ legal layouts, f_s retrieval is biased by lighting and viewpoint—it picks the *appearance-closest* image, not the *layout-closest* one—and parts that are individually normal are then declared anomalous because they fail to match the wrong template.

(F2) Free-recombination assumption. The complementary failure appears if one instead pools all parts from all K support images into one memory bank (the UniVAD [2] route). Pooling borrows parts *across layouts*: a genuinely anomalous product whose parts happen to be drawable from different legal layouts gets "balanced" by parts from another mode, and is scored normal. The failure is measurable: UniVAD reports only 62.0 LOCO image-AUROC at 1-shot, precisely the regime where unconstrained pooling is left naked.

Neither extreme is right. The correct question is not "match one image or pool everything"; it is *which factors may be recombined, and which may not*. Recombining parts *within a legal layout mode* is allowed; borrowing parts *across modes* is not; and whether a pairwise relation is even stable enough to constrain assembly must be decided by the support images themselves, via leave-one-out agreement.

Our approach: mode-conditional assembly. Logical anomaly detection sits at an awkward point: full-shot methods that explicitly model composition maps [13] or decompose detection into patch, component, and global experts [18] need hundreds of training images, while the strongest few-shot competitors either greedily retrieve a single nearest normal reference image and match objects onto it [14], or pool all normal components into one flat memory bank [2]. Neither option tolerates the *multi-layout normality* that real products exhibit—a handful of support images contain several equally legal arrangements, each with its own parts inventory and its own stable pairwise relations. MCA (**M**ode-**C**onditional **A**ssembly; framework: *compositional set inference at* $K \leq 4$) treats the K support images as an unordered set of samples from an unknown number of layout modes, and builds no neural network. At $K \leq 4$ it enumerates all Bell(K)=15 partitions of the support set [7] and scores each partition by *leave-one-image predictive NLL* [36, 38]: hold out one support image, estimate a mode from the remaining images, and ask how well those images predict the held-out one. Partitions within ± 1 nats of the best are retained (an Occam's window [30]) and averaged by pseudo-BMA weights, so the arbitrariness of near-tied partitions does not propagate to query scores. Inside each mode, a query is scored by conjugate posterior-predictive assembly cost [16, 32]:

- a *parts-inventory* term, a Gamma–Poisson negative-binomial (NB) posterior-predictive NLL counting whether each legal part type appears the right number of times;
- a *pairwise-relation* term, a per-dimension 1D Normal–Inverse–Gamma (NIG) Student-t posterior-predictive NLL penalizing part offsets inconsistent with the mode's measured layout;
- a leave-one-support-out (LOSO) *soft gate* $s_{ab} \in [0, 1]$ that mutes relation types confirmed by fewer than two held-out agreements, so weak rules never fire.

Because every term is a posterior predictive rather than a plug-in estimate, weak evidence (few observed images, few observed relations) automatically widens the predictive distribution and shrinks the penalty—the algorithm inherits "weak rules must not false-alarm" from conjugacy, with no hand-tuned weights. On single-mode episodes the mode selector is the identity and MCA degenerates to nearest-support matching: a bounded fallback that cannot undercut the existing lower bound.

What we claim. Our claims are deliberately narrow:

1. **Problem formulation, not tool-stacking.** We adopt standard closed-form Bayesian tools (conjugate posterior predictive distributions, exact leave-one-out scoring, model averaging); our contribution is the *formulation* of class-agnostic few-shot logical anomaly detection as set inference over latent layout partitions, plus the small-sample regime in which it closes exactly, with no EM and no initialization.
2. **Small-sample identifiability as evidence.** Relations are unidentifiable at $K = 1$ (each relation has at most one

observation) and become usable at $K = 4$; we report the gain-vs-K curve and a de-circularized synthetic benchmark whose three controls must pass for the mechanism claim to hold.

3. **Honest falsifiability.** We write our kill-lines into the experiment design: if unconstrained pooling is not worse than MCA, if gains survive removing the mode selector or the relation term, or if the paired category-level CI crosses zero with the gain below 1 p.p., we report that the mechanism claim does not hold.

Results. [**[TODO: Main table pending; full 5-class × 10-episode runs scheduled ETA 2026-10-30. Pilot single-episode numbers are reported only as implementation-fidelity diagnostics, §4.2.]**]

2. Related Work

[**[TODO: Numbers below are taken from the cited papers' own tables; we re-run no competitor. Bibliographic gaps (author lists) are tracked in TODO_refs.md.]**]

2.1. Few- and zero-shot anomaly detection.

Two sub-lines saturate the structural regime. The *text-prompt line* starts from WinCLIP [23] (thousands of hand-designed CLIP text templates), moves to learned prompts with only normal samples [29], object-agnostic prompt learning [42], hybrid learnable prompts [11], and mutual scoring of unlabeled images [28]. These methods require category names in the prompts or auxiliary normal/abnormal supervision, and they score local appearance patches, not compositions. The *vision-only line* drops text: DINOv2 patch features turn few-shot AD into retrieval (AnomalyDINO [12]: 96.6% one-shot MVTec-AD), fusion of multi-layer foundation features pushes to 97.4% [39], and SubspaceAD [27] shows a single PCA subspace over DINOv2 patch features suffices (97.1% one-shot MVTec-AD, no training). Crucially, SubspaceAD's appendix concedes that patch-based methods "focus on local appearance and therefore do not explicitly account for logical or structural anomalies, such as missing or misplaced components"—the exact gap we target. We deliberately do not enter that saturated table; our structural branch is a frozen-feature residual kept out of the main claim, and appears only in an auxiliary table (§4.6).

2.2. Logical and compositional anomaly detection.

MVTec LOCO [8] defines logical anomalies as missing / extra / position errors with pixel annotations over five industrial categories; CAD-SD [21] (776 images, screw rods with hex nuts at random angles) adds a co-occurrence relation dataset. *Full-shot systems* operate per category with hundreds of normal images: SALAD [13] learns a composition map (cluster DINO features into $K = 6$ groups, train a UNet composition discriminator) and reaches 96.1 LOCO image-AUROC; CSAD [20] segments components for logical anomaly detection; PSAD [37] adds prompting-based self-attention. *Few-shot systems* are the relevant track: ObjectCore [14] is the direct competitor—a prompt-"part" detector, per-category fine-tuned GroundingDINO with a SAM mask pool, an appearance-and-mask-area Hungarian matching cost, and a structural PatchCore branch (no coreset subsampling). Its 4-shot LOCO ablations quantify the pieces: removing fine-tuning -11.8 p.p., only the structural branch -12.1, only the logical branch -10.2, restoring box-regression loss -3.2, Swin-B→T backbone -4.5, SAM patch features -3.3, mask→box matching -4.1. Its logical memory stores, per support image, a global feature f_s and a part set O_s; at test time it retrieves the *single* O_s nearest to f_s and matches against only that image—it has no notion of layout modes and no pairwise relation term. AnomalyMoE [18] decomposes detection into patch/component/global experts but needs its generalist training regime (one-shot LOCO 67.9%). Training-free VLM-prompt systems—LogSAD [5], LogicQA [4], LogicAD [3]—keep category or defect words in the prompt and are out of our class-agnostic track. AnomalySD [1] uses Stable-Diffusion inpainting as a *generative* residual corrector; we borrow edit attribution only as an unscored residual explanation (§3). *Sentinel:* the same group's newest preprint, PyramidCore [15] (MELECON 2026), adds feature pyramids to the object-memory idea; its numbers are not yet public, so we claim no differentiation against it and re-run the comparison before submission.

2.3. Structured and relational modeling.

Relation-based anomaly scoring exists outside the few-shot industrial track. UniVAD [2] flattens all support components into a component graph and pools over them—the unconstrained pooling we empirically attribute to cross-mode borrowing (LOCO 1-shot 62.0). The Hypergraph Normal World Model [6] computes pairwise $\cos(z_u, z_v)$ energies on a fixed DINOv2 grid with a single normal world and no mode discovery. LayoutAD [41] reasons about semantic-geometric misalignment at the COCO scene level. None groups $K \leq 4$ support images into layout modes, conditions an assembly cost on the selected mode, or uses leave-one-support-out relation confirmation. Methodologically, our assembly cost has the standard unary-plus-pairwise energy form of CRFs [24, 26] and graph-matching assignment energies [10, 17]; the rectangular matching used in our row-1 baseline is the classical Hungarian algorithm [9, 25, 31]. What is new is not the energy form but the small-sample inference: at $K \leq 4$ we enumerate all Bell(K) support-set partitions [7] instead of running the classical mixture

Table 1. Positioning on the class-agnostic few-shot logical AD track. FS=few-shot; Mode=layout-mode discovery from the support set; Rel=mode-conditional pairwise-relation scoring; LOIOO=leave-one-support-out confirmation.

System	FS	Mode	Rel	LOIOO
ObjectCore [14]	✓	—	—	—
SALAD [13]	(full-shot)	—	(param.)	—
UniVAD [2]	✓	—	—	—
Hypergraph NWM [6]	subset	—	fixed-grid	—
AnomalySD [1]	✓	—	—	—
MCA (ours)	✓	✓	✓	✓

machinery (EM / reversible-jump MCMC [33]; early drafts used BIC [35]), and we score relations through conjugate posterior predictives gated by leave-one-out agreement. Table 1 places MCA against the systems we most overlap.

3. Method

Setup. A category c exposes a support set $S = \{s_1, \dots, s_K\}$, $K \in \{1, 2, 4\}$, of normal RGB images; no category name is given, and every image is grounded with the single generic prompt "part". A test image x is scored by how plausibly its parts assemble as a legal composition. MCA is non-parametric: nothing is learned by gradient descent.

Scope of the method. Every numerical device in this paper is a standard, closed-form statistical tool: exact leave-one-out predictive scoring [36, 38], conjugate posterior-predictive distributions [16, 32], and model averaging within an Occam window [19, 30]. Our contribution is not any single device but the *inference problem*—at $K \le 4$, class-agnostic logical anomaly detection becomes a compositional set inference over latent layout partitions, which closes exactly by enumeration rather than by EM or MCMC—and the small-sample regime in which that closure is identifiable. The assembly cost itself has the familiar unary-plus-pairwise energy form of structured prediction [17, 26]; what is unusual is that the pairwise term is a mode-conditional, leave-one-out-gated posterior-predictive NLL rather than a discriminative edge potential.

3.1. Slot extraction (shared by all compared rows).

We follow the exact slot pipeline of ObjectCore [14] so that our ablations differ only in how slots are consumed. The rectangular matching used inside the row-1 baseline is the classical Hungarian algorithm, padded to a square matrix with its maximum cost [9, 25, 31]. GroundingDINO-Swin-B is fine-tuned per category on its K support images under the published recipe (100 epochs, AdamW, lr 10^{-5}, batch 1, detection confidence 0.25, 800×800 input, ~ 5 min per category), using only the GLIP classification loss $\mathcal{L}_{\text{cls}}$ and deliberately *omitting the box-regression loss* $\mathcal{L}_{\text{bbox}}$ (restoring it costs -3.2 p.p.). SAM-ViT-B produces masks inside the boxes, and each image yields a slot set $X_i = \{x_1, \dots, x_n\}$ with:

- appearance $z_i \in \mathbb{R}^{d_z}$: frozen Swin three-layer patch features (after 5×5 mean filtering, PatchCore-style), average-pooled inside the mask, ℓ_2-normalized;
- geometry $g_i = (u_i, v_i, w_i, h_i)$: mask bounding box, normalized to the image;
- one global key f_s: the image-average patch feature, used *only* by the row-1 retrieval baseline.

3.2. Part types and relation samples.

Part types. Slots from all K support images are clustered once by single-linkage union-find (cosine ≥ 0.9, merging in descending similarity, at most 10 prototypes); the type prototype is $\mu_c = \ell_2(\bar{z}_c)$. Type prototypes are estimated once per episode and are *not* re-clustered when a support image is held out for leave-one-out evaluation.

Relation samples. For an ordered type pair (a, b) we use a canonical anchor (the smaller type index; ties broken by slot order) and collect, for each co-occurrence,

$$q = (q_1, q_2, q_3, q_4) = \left(\Delta u,\ \Delta v,\ \log \tfrac{w'}{w},\ \log \tfrac{h'}{h}\right), \quad (1)$$

the offset and log-size ratios of the second part relative to the anchor. The prior scale Σ_0 is data-driven: per-dimension MAD$\times 1.482^2$ (robust scale, [34]) over all relation samples of the episode; dimensions with MAD zero (frequent at $K = 4$) fall back to the diagonal $(0.09, 0.09, 0.04, 0.04)$.

3.3. Mode selection: leave-one-image predictive NLL.

A *partition* π of the K support images is a candidate layout hypothesis: images in the same block share a mode. With $K \le 4$ there are only Bell(4)$= 15$ partitions [7], which we enumerate exactly (no EM, no initialization). At larger n the classical treatment of an unknown mode count would be EM or reversible-jump MCMC [33] (an earlier draft of ours used BIC [35]); at $n = 4$ enumeration is cheaper and exact. For partition π, estimate each block's posterior parameters from its images *except* image s_k, then predict s_k:

$$L_{\text{pred}}(\pi) = \sum_{k=1}^{K} \text{NLL}_{\text{pred}}\left(s_k \mid \theta^{(-k)}_{m(k)}\right), \quad (2)$$

where NLL_{pred} is the sum of the three posterior-predictive terms (§3.4) in nats, *without* the relation weight w_{rel}. An empty block ($N_m = 0, t = 0$) predicts with the prior parameters, so its relation cost is 0 rather than $+\infty$ (Eq. (3.4)); over-splitting is self-penalized only through the lost peer information in the count term. We keep all partitions within

a near-neighbour window—an Occam's window [30] in predictive-NLL units (a 1-nat threshold corresponds to an e-fold likelihood ratio):

$$\Pi_{\text{win}} = \{\pi : L_{\text{pred}}(\pi) \le \min_{\pi'} L_{\text{pred}}(\pi') + 1 \text{ nats}\}. \quad (3)$$

With $n = 4$ we do not claim to recover the true mode count; we claim only a predictive-level model choice.

3.4. Within-mode shrinkage and posterior prediction.

Count term (Gamma–Poisson → NB). With Gamma($\alpha_0 = 1, \beta_0 = 1$) prior, a mode m with N_m images and total count $n_{\text{tot}}^{(c)} = \sum_{k \in m} h_k(c)$ of part type c has posterior $\alpha_n = 1 + n_{\text{tot}}^{(c)}$, $\beta_n = 1 + N_m$, and posterior-predictive count $\sim \text{NB}(r = \alpha_n,\ p = \beta_n/(\beta_n + 1))$. Its closed-form NLL is

$$\ell_{\text{NB}}(n) = \gamma(\alpha_n) + \gamma(n+1) - \gamma(\alpha_n + n) + (\alpha_n + n)\log(\beta_n + 1) - \alpha_n \log \beta_n, \quad (4)$$

where $\gamma(\cdot) = \log\Gamma(\cdot)$. A missing part ($n = 0$) costs $\alpha_n \log(1 + 1/\beta_n) > 0$; a type enters the inventory only when the posterior mean $\alpha_n/\beta_n \ge 0.5$; per-type contributions are winsorized at $3\times$ their natural scale (detected misses are not protected). $\gamma(n+1)$ is invariant across modes and cancels in the marginalization below.

Relation term (per-dimension 1D NIG → Student-t). The updates are the standard conjugate Gaussian analysis [16, 32]; μ_n shrinks the empirical mean toward the prior 0 by the convex shrinkage factor $t/(1+t)$ [22]. For a type pair observed t times in mode m with empirical mean $\mu_{\text{emp},j}$ and empirical scatter $S_{\text{emp},j} = \sum_i (q_{ij} - \mu_{\text{emp},j})^2$ in dimension $j \in \{1, \dots, 4\}$:

$$\kappa_n = 1 + t, \quad \nu_n = 1 + t, \quad \mu_{n,j} = \frac{t\,\mu_{\text{emp},j}}{\kappa_n}, \quad (5)$$

$$S_{n,j} = S_{0,j} + S_{\text{emp},j} + \frac{t}{\kappa_n}\mu_{\text{emp},j}^2, \quad (6)$$

$$\sigma_{\text{pred},j}^2 = S_{n,j}\,\frac{\kappa_n + 1}{\kappa_n\,\nu_n} \quad (\text{floor } 10^{-12}) \quad (7)$$

The predictive distribution factors as $q_{\text{new}} \sim \prod_{j=1}^{4} t_{\nu_n}(\mu_{n,j}, \sigma_{\text{pred},j}^2)$, and its NLL is

$$\ell_t(q) = \sum_{j=1}^{4} \Big[\gamma(\tfrac{\nu_n}{2}) - \gamma(\tfrac{\nu_n+1}{2}) + \tfrac{1}{2}\log(\pi\nu_n\sigma_{\text{pred},j}^2) + \tfrac{\nu_n+1}{2}\log\big(1 + \tfrac{(q_j - \mu_{n,j})^2}{\nu_n\sigma_{\text{pred},j}^2}\big)\Big] \quad (8)$$

Every $t \ge 0$ is defined ($t = 0$ gives a heavy df$= 1$ tail; $t = 2$ gives df$= 3$); the per-dimension NIG avoids the improper prior of a full-dimension NIW at $\nu_0 = 1$.

Appearance term. $1 - \cos(z_i, \mu_{c(i)})$ per slot; a slot whose best prototype has $\max_c \cos < 0.5$ is novel and pays a fixed 1.0 nats.

Empty mode. $N_m = 0, t = 0$: $\alpha_n = \alpha_0$, $\beta_n = \beta_0$, $\mu_n = 0$, $\sigma_{\text{pred}}^2 = \frac{\kappa_0+1}{\kappa_0} S_0$, and the soft gate below gives $s_{ab} = 0$, i.e. $R_t = 0$—*not* $+\infty$.

3.5. LOSO soft gating of weak relations.

A type pair contributes its relation cost only when leave-one-support-out (LOSO) agreement supports it. With $t \le 1$ observations $s_{ab} = 0$. With $t \ge 2$ we hold out one observation per fold and count the fraction of held-out samples falling inside $\mu_n \pm 2.0\sqrt{\text{diag}\,\Sigma}$; we also force $s_{ab} = 0$ when $\text{trace}(\Sigma_{\text{emp}}) > 2\,\text{trace}(\Sigma_0)$ (the parts are exchangeable within the mode). The gate is multiplicative, $s_{ab} \in [0, 1]$.

3.6. Query scoring: two-level pseudo-BMA.

For a query x, mode m of partition π scores

$$C(x \mid m, \pi) = D_{\text{NB+cos}}(x \mid m) + w_{\text{rel}}\,R_t(x \mid m), \quad (9)$$

where $D_{\text{NB+cos}}$ sums Eqs. (4) over present types and the appearance term over slots, and $R_t = \sum_{i<j} s_{ab}\ell_t(q_{ij})$ sums gated relation NLLs. We average over the window with a *two-level* normalization—first over partitions, then uniformly within each partition:

$$\boxed{s_{\text{logic}}(x) = -\log \sum_{\pi \in \Pi_{\text{win}}} p(\pi)\,\frac{1}{|\pi|}\sum_{m \in \pi} \exp\big(-C(x \mid m, \pi)\big),} \quad (10)$$

$$p(\pi) = \frac{\exp(-L_{\text{pred}}(\pi))}{\sum_{\pi' \in \Pi_{\text{win}}} \exp(-L_{\text{pred}}(\pi'))}. \quad (11)$$

We call $p(\pi)$ *pseudo-BMA weights*: they are proportional to $\exp(\text{elpd}_{\text{loo}}(\pi))$ from exact leave-one-out predictive log-scores [38, 40], *not* marginal-likelihood posterior model probabilities, and we report them as such (see §6). We never use the joint weight $|\pi|\,p(\pi)$: that would reward partitions with more blocks and destroy the anti-over-splitting property of the leave-one-out criterion. Inner and outer sums use max-shift log-sum-exp in float64. Because L_{pred} contains no w_{rel}, mode selection is *invariant* to w_{rel} by construction; w_{rel} acts only at query scoring. The single scalar w_{rel} is calibrated once globally on two held-out pilot categories (half split: synthetic part-deletion / cross-mode transplant on half, evaluation on the other half) by maximizing the anomaly–normal score margin over $\{0.1, 0.3, 1, 3, 10\}$, frozen afterwards, and robustness is reported by leaving out one calibration category.

3.7. Residual editing attribution (not scored).

On residuals with Mahalanobis distance > 3, we test whether a single deletion / move / swap explains $> 80\%$ of the residual. This is attribution only; it does not enter s_{logic}, following AnomalySD's [1] prior on generative repair.

3.8. Structural score and fusion.

A frozen DINOv2 patch residual plus the existing LDM scratch/dent branch gives a structural map A_S (no coreset subsampling, matching the row-1 baseline), reported *separately*. Fusion is isomorphic to ObjectCore: Gaussian smoothing $\sigma = 4$, top-1% mean pooling of $A = A_S + A_L$. Every main number is reported twice: pure-logic column and fused column; gains must hold in the pure-logic column.

3.9. Propositions and the identifiability curve.

Prop. 1 (degeneration, empirical). On $M = 1$ episodes (optimal partition has one block), MCA and row-1 nearest-support matching rank test images nearly identically: ΔAUROC$\leq$ 1 p.p. with Spearman rank correlation reported on the $M = 1$ bucket. We state this as a rank property, not a scalar bound (nats and top-1% pixel pooling are not commensurable).

Prop. 2 (pooling risk, expectation form). If a relation $q \sim \mathcal{N}(\mu_{m'}, \Sigma_{m'})$ is scored under mode m,

$$\mathbb{E}_{q\sim m'}\big[(q-\mu_m)^\top \Sigma_m^{-1}(q-\mu_m)\big] = \Delta^\top \Sigma_m^{-1}\Delta + \mathrm{tr}(\Sigma_m^{-1}\Sigma_{m'}), \quad (12)$$

i.e. the expected relation penalty is the two-Gaussian KL (up to constants), gated multiplicatively by s_{ab}. With $t \leq 3$ the estimate of Σ_m is imprecise; the claim is qualitative, and no quantitative bound is asserted.

Prop. 3 (conjugate pessimism, holds for the algorithm). The posterior predictive is wider than the empirical distribution: the 1D-NIG predictive variance carries two inflation factors, $1/\nu_n$ and $(\kappa_n+1)/\kappa_n$, and the NB posterior-predictive variance is the mean times $(1 + 1/\beta_n)$, where $\beta_n = 1 + N_m$—fewer support images in the mode, wider the count prediction. Because our algorithm scores with the predictive (not plug-in), weak evidence (small t or n) automatically widens the predictive and shrinks the penalty—"weak rules must not false-alarm" follows from conjugacy rather than tuning.

Identifiability curve. At $K = 1, t \leq 1$ so $s_{ab} \equiv 0$ and the relation term is off by construction; at $K = 2$ relations are weak; at $K = 4$, df$= 3$ Student-t posteriors and LOSO voting can activate. The measured gain-vs-K curve is the identifiability evidence; a de-circularized synthetic benchmark (§4.4) provides causal confirmation.

4. Experiments

4.1. Datasets and protocol.

MVTec LOCO [8]: five categories (breakfast_box, juice_bottle, pushpins, screw_bag, splicing_connectors), 3644 images, logical anomalies = missing / extra / position errors with pixel annotations. **CAD-SD** [21]: one category (screw rods + hex nuts), 776 images at random angles, co-occurrence anomalies; the few-shot protocol is the one defined by ObjectCore [14]. We follow the class-agnostic protocol exactly: prompt "part"; per-category GroundingDINO fine-tuning on the K support images (100 epochs, AdamW, lr 10^{-5}, batch 1, detection confidence 0.25, 800×800, $\sim$ 5 min/category), using only the GLIP classification loss $\mathcal{L}_{\mathrm{cls}}$ and omitting the box-regression loss $\mathcal{L}_{\mathrm{bbox}}$, as published. The Hungarian cost matrix is padded to a square with its maximum distance; the logical anomaly map weights matched-pair costs and unmatched slots by their mask area. Support sets are sampled randomly; four rows share the *same* 4-shot sampling. ObjectCore's official code was not publicly available at writing time (its repository link returned 404); row 1 is our faithful reimplementation from Eqs. 1–7 of [14], accepted only inside the band 80.8±2.0. All main numbers below are image-level AUROC, macro-averaged over the five LOCO categories after per-category scoring; we additionally report the 3-repeat protocol of [14] alongside our primary 10-episode protocol.

Table 2. Class-agnostic 4-shot image-AUROC on MVTec LOCO (macro over 5 categories; 10 episodes/category, shared support sampling; per-category GroundingDINO fine-tuning as in [14]). Row 1 reproduced from Eq. 1–7 of [14] (official code unavailable). Data-policy TBDs: filled only with measured numbers; no extrapolation.

Row	Method	Pure logic
1	Nearest single support (ObjectCore reimpl.)	**[TODO: TBD ETA 10-3**
2	Unconstrained pool	**[TODO: TBD ETA 10-3**
3	**MCA (ours)**	**[TODO: TBD ETA 10-3**
4	Row 3 + edit attribution	**[TODO: TBD ETA 10-3**

4.2. Main comparison: four rows, two columns.

Table 2 reports the pure-logic column and the fused (pure-logic + structural) column for: (1) nearest-single-support matching (ObjectCore reimplementation); (2) unconstrained pooling of all support parts; (3) MCA; (4) MCA + single-edit residual attribution. [**[TODO: Rows 3–4 pending full runs, ETA 2026-10-30.]**]

Pilot diagnostics (not a main-table number). At an early implementation-fidelity stage, the single-episode, No-fine-tuning, logic-only breakfast-box diagnostic measured: row 1 0.709 (three-layer multi-scale features 0.680), row 2 0.706, and +A_S fusion 0.720; a first full fine-tuning attempt collapsed to 0.628 (diagnosed as over-fitting on four pseudo-boxes, under repair). These are *single-episode, No-FT, scalar-sum* numbers—they sit $\sim$11.8 p.p. below the fine-tuned detector (ObjectCore's own No-FT ablation on LOCO 4-shot) and lack the $\sigma = 4$ anomaly-map/top-1% aggregation, so they are not comparable to 80.8 and are reported only to evidence that

Table 3. Ablations (LOCO 4-shot, pure-logic column, macro AUROC). Each row maps to a kill-line; gain should *disappear* after A1/A2 for the mechanism claim to survive.

ID	Setting	ΔAUROC vs. MCA	Kill-line
A1	no mode selection	[TODO: TBD]	gain must vanish
A2	no relation term	[TODO: TBD]	gain must vanish
A3	no edit attribution	[TODO: TBD]	editing must not add
A4	detector fairness	[TODO: TBD]	no gain from detector
A5	LOSO gate off	[TODO: TBD]	FP should rise
A6	w_{rel} grid	[TODO: TBD]	sensitivity
A7	plug-in vs. postpred	[TODO: TBD]	Prop. 3 evidence
A8	uniform vs. loo prior	[TODO: TBD]	> 0.5 p.p. upgrades story
A9'	w_{rel} rank sens.	[TODO: TBD]	Spearman

the slot→Hungarian→unmatched-area chain is directionally consistent with ObjectCore's "only M_L" anchor (70.6).

4.3. Ablations tied to the mechanism claims.

Table 3 lists the causal ablations, each mapped to a kill-line: A1 remove mode selection (force $M = 1$); A2 remove the relation term ($R = 0$); A3 row 4 vs. row 3 (if editing lifts AUROC, the mechanism is editing, not assembly); A4 detector fairness (four rows share slots; a stronger-detector control); A5 LOSO gating off (weak rules still score—false positives should rise); A6 w_{rel} grid $\{0.1, 0.3, 1, 3, 10\}$; A7 posterior-predictive vs. plug-in scoring (Prop. 3 experimental); A8 uniform vs. leave-one-out BMA weights; A9' rank-sensitivity of query scores to the w_{rel} grid (Spearman, median/90th percentile). [**[TODO: numbers pending, ETA 11-01.]**]

4.4. Analysis and de-circularized synthesis.

R1 multi-mode concentration: per-episode selected mode count M vs. row3−row1 gain (the mechanism should concentrate on $M \geq 2$ episodes and be ≈ 0 on the $M = 1$ bucket). **R2** confirmed-relation-count buckets. **R2s** identifiability curve: relation-term gain at $K = 1, 2, 4$ ($K = 1 \approx 0$). **R3** degeneration report: per-class M distribution, single-linkage merge collapse, MAD fallback trigger rate. **R4** edit attribution statistics aligned to LOCO defect subtypes. **R7** subtype AUROC (missing / extra / position). **R8** injected-detector-miss sensitivity (winsorization on/off).

Synthetic, de-circularized. We place real part features into known geometric layouts (μ_m, Σ_m known, not grouped by our own features) and inject anomalies from a distribution *different* from our detection statistic (3σ offsets, structural rearrangements, not $\pm 2\sigma$ perturbations). Three controls must pass: (i) on $M = 1$ synthesis, row 3≈row 1; (ii) on multi-mode synthesis, row 2 (pooling) must fail; (iii) on wide-Σ synthesis, the relation term contribution must be ≈ 0. Failure demotes this experiment to supplementary. [**[TODO: numbers pending, ETA 11-03.]**]

4.5. CAD-SD as a degeneration check.

CAD-SD has one category at random angles, so the relation term is off by construction ($s_{ab} \approx 0$); we report only a point estimate against ObjectCore's 96.5±0.6 (3-repeat std ≈ 0 is itself reported in [14]), and make no significance claim. If row 3 > row 2 here, the effect must be attributed to the inventory co-occurrence (screw/nut counts); we do not claim to beat 96.5.

4.6. Auxiliary tables.

Structural-branch sanity on MVTec AD / VisA: reported without SOTA claims. [**[TODO: TBD.]**]

4.7. Statistical protocol and written-in kill-lines.

Per-category AUROC is computed on that category's normal+logical split, then macro-averaged over the five categories; the unit of resampling is the *category*, not the episode. The paired difference $d_c = \text{AUROC}_3(c) - \text{AUROC}_1(c)$ ($c = 1..5$) is reported with a paired t interval (df= 4) and a 10,000-replicate category bootstrap. The mechanism claim requires: point estimate ≥ 1 p.p. *and* CI lower bound > 0; a positive mean with CI crossing zero is reported as a trend, not a result. We write the following kill-lines into the design: on 4-shot class-agnostic LOCO, row 1 must fall in $[78.8, 82.8]$ and row 3 must exceed 80.8; on CAD-SD we only point-estimate against 96.5±0.6. If unconstrained pooling (row 2) is not worse than row 3, or if the gain survives A1/A2, or if the gain is below 1 p.p. with the paired CI crossing zero, the mechanism claim does not hold and we report it as such.

5. Conclusion

[**[TODO: To be finalized with frozen numbers. Planned text:]**]We reframed class-agnostic few-shot logical anomaly detection as compositional set inference at $K \leq 4$: enumerate the 15 layout partitions of the support set, select and average them by leave-one-image predictive NLL, and score a query by conjugate posterior-predictive assembly cost. The design makes three predictions that the experiments are built to falsify—single-mode episodes degenerate to nearest-support matching, unconstrained pooling is punished, and weak relations are muted by their own posterior predictive. [Fill measured outcomes.]

6. Limitations and Failure Modes

We list, as designed rather than after the fact, the known limitations:

- $K = 4$ **statistical power.** A candidate relation is confirmed by leave-one-support-out votes, i.e. at most $K -$

$1 = 3$ observations per mode; when support splits into two modes each relation has ~ 2 images and df$= 3$ Student-t tails are heavy. We mitigate with soft voting and the K-curve, but the effect size is intrinsically underpowered and we do not hide it.

- **Detector dependence.** Everything hinges on GroundingDINO-SAM slots; a missed detection creates a false "missing part". We use dual confidence, winsorized inventory terms, and an injected-miss study (R8).
- **CAD-SD random angles.** Rotating parts destroy anchored offsets, so the relation term is off by construction there; CAD-SD is a degeneration check, not a lead benchmark.
- **ObjectCore code unavailable.** Row 1 is a faithful reimplementation from the published equations, not the authors' code; if the authors release it, we will re-run row 1 and report both.
- **Small-sample identifiability.** At $K = 1$ relations are unidentifiable; the entire relation contribution is an emergent phenomenon of $K = 4$, and the de-circularized synthetic controls are load-bearing for the causal claim.
- **Calibration cost.** The single scalar w_{rel} is tuned on two held-out pilot categories; robustness is reported by leave-one-calibration-out.
- **Pseudo-BMA underestimates uncertainty.** Partition weights are pseudo-BMA weights from exact LOO elpds, not posterior model probabilities; pseudo-BMA is known to under-weight model uncertainty relative to stacking [38, 40]. At $n = 4$ with an exact leave-one-out this is acceptable, but partition-choice variance is not propagated into the category-level CIs below.

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Supplementary Material (outline)

[**[TODO: Sections to fill during experiment waits.]**]

- S1: Full pipeline figure (slot extraction → type clustering → relation collection → 15-partition leave-one-out → two-level BMA).
- S2: Hyperparameter table (all constants, MAD fallback rates).
- S3: Per-class AUROC breakdown and mode-count M distributions.
- S4: Full ablation grid numbers; w_{rel} sensitivity curve.
- S5: De-circularized synthetic benchmark details and three controls.
- S6: Failure cases (missed slots, repetitive texture, viewpoint drift).
- S7: Reproduction checklist (support seeds, fine-tuning recipe, environment).